LOAN GROWTH MODEL

MARCH 2024

Prepared by:
Strategic Planning Department
8 MARCH 2024



Agenda

1. Introduction
2. Factors
3. Correlation Analysis
4. Multivariate Analysis
5. Projection
6. Conclusion

Introduction

EXIM Bank's Yearly Performance

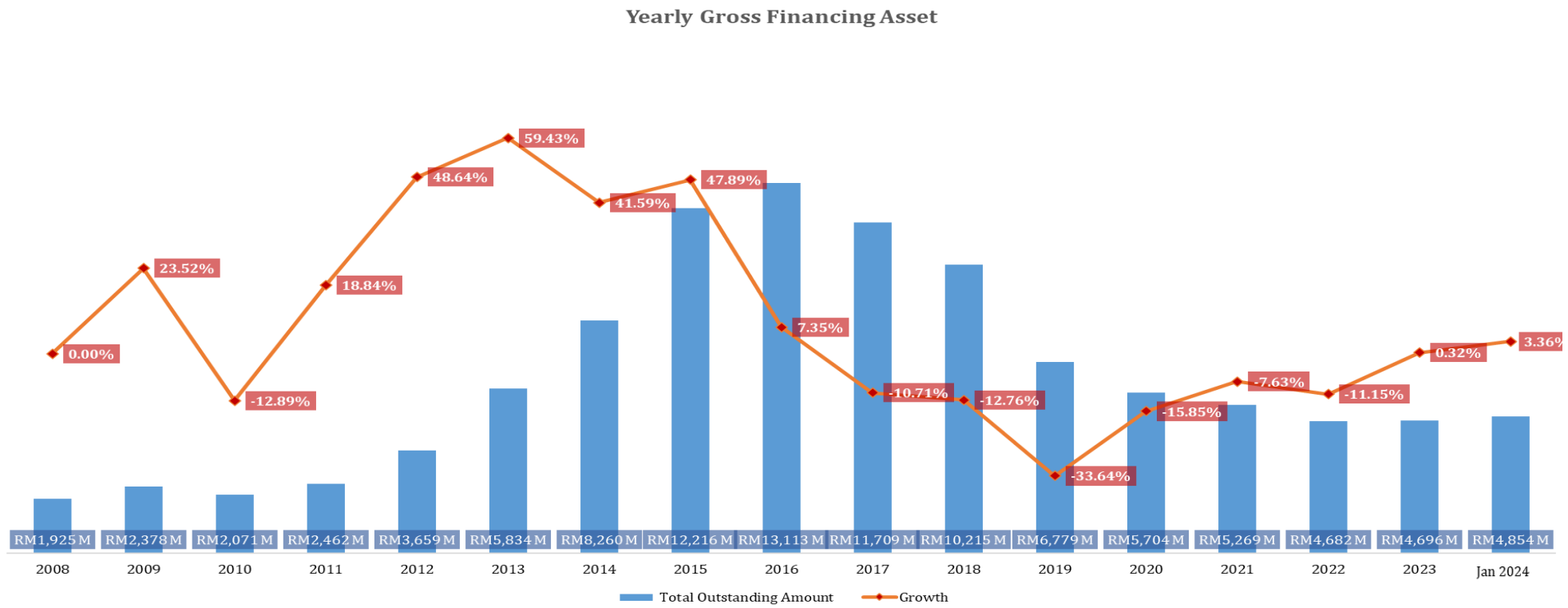
The figure shows the EXIM bank borrowing asset performance, while below shows the Measures of Central Tendency of the Yearly Gross Financing Asset Performance:

- **CAGR** Gross financing asset of **5.59%**
- **Minimum** total borrowing of **RM 1,925 M** & **-33.64%** growth
- **Maximum** total borrowing of **RM 13,113 M** & **59.43%** growth
- **Average** total borrowing of **RM 6,225 M** Quarterly & **9.14%** growth
- **Standard deviation** total borrowing of **RM 3,649 M** & **27.77**% growth

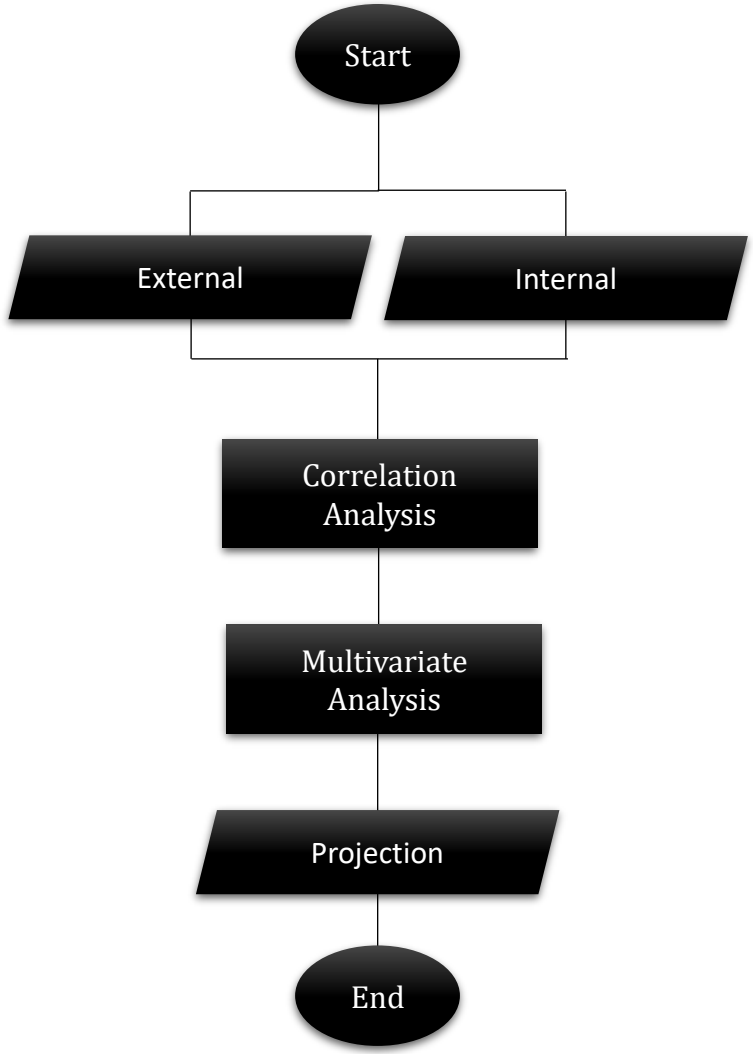
Research Questions

Based on the performance below, the research question is as follows:

- What are the main factors that driven loan growth in EXIM bank?



Process Flow



Factors

Internal

Definition

Disbursed Account

- Measures the capability of EXIM Bank to disburse loan.

No. of Application

- Number of application pending to become EXIM customer.

Staff Strength

- Number of staff working in EXIM Bank by front office and back office on day-to-day operation.

Justification

Disbursed Account: EXIM Bank's capability to disbursed loan will increase the gross financing asset.

No. of Application: Indicate the possibility of maximum potential customer EXIM Bank can do business.

Staff Strength: Increase in staff will possibly leads to better performance.

External

Definition

US Dollar Index

- Measures the value of the US dollar against a basket of six major currencies: the euro (EUR), Japanese yen (JPY), British pound (GBP), Canadian dollar (CAD), Swedish krona (SEK), and Swiss franc (CHF).

FX (USD to MYR)

- The exchange rate between the US Dollar and Malaysian Ringgit, impacting trade, investment, and economic stability in Malaysia

OPR

- Overnight Policy Rate is interest rate set by the central bank impacting borrowing and lending costs, influencing the economy..

GDP (MY & US)

- Measures the total market value of all final goods and services produced within its border during a specific period.

Fed Funds Target

- The interest rate for US Banks lending cash set by the Federal Reserve that influences the entire economy.

FTSE Bursa Malaysia KLCI

- The benchmark stock index for Malaysia.

Justification

USD Dollar Index: Weaking USD makes borrowing USD cheaper, potentially loan growth by lower interest rates

FX: Appreciation of RM reduces of foreign currency-denominated debt, potentially allowing for loan growth

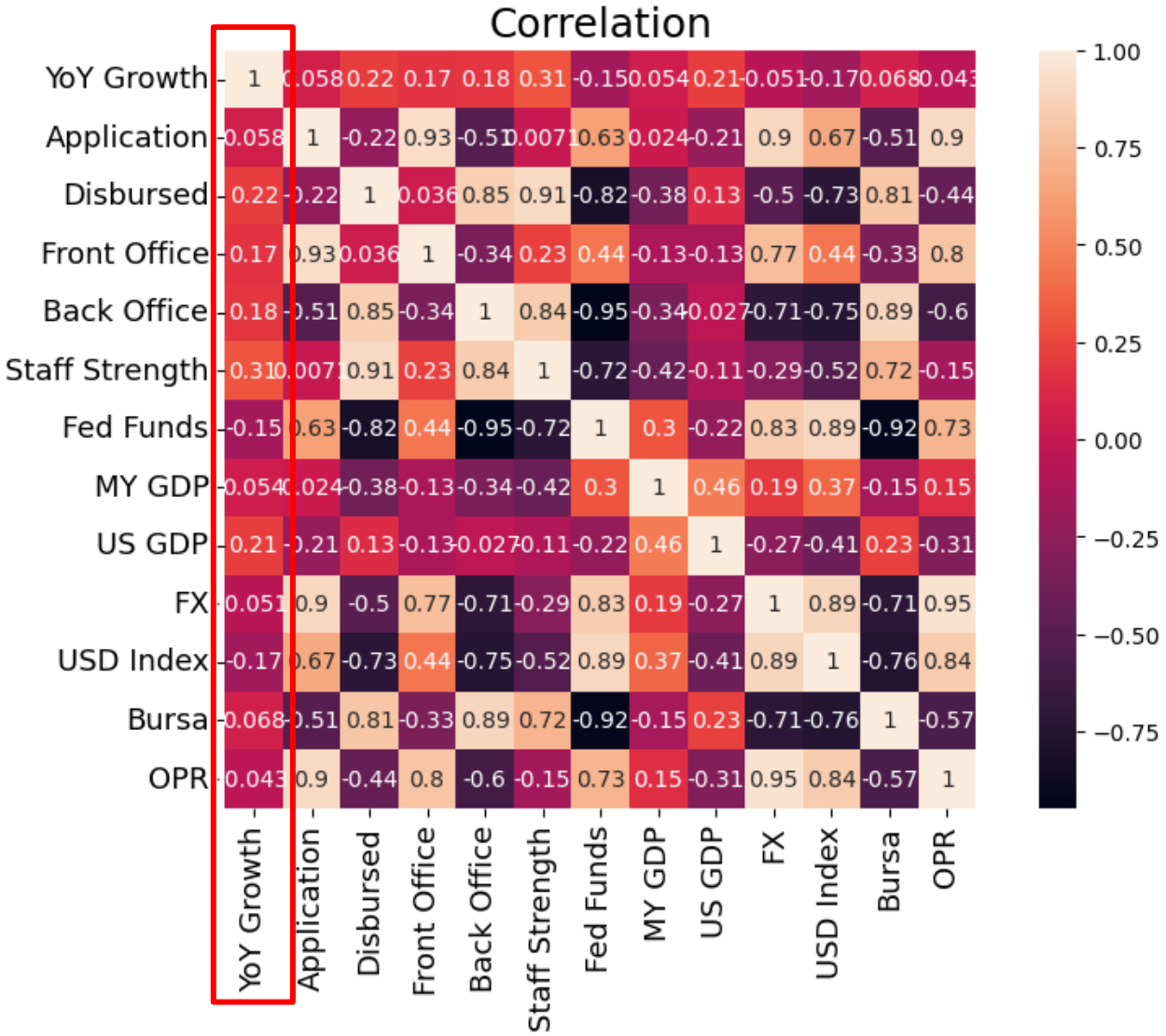
OPR: OPR increase makes borrowing more expensive due to increased of cost of fund.

GDP: Stronger GDP indicates a growing economy with potentially higher demand for financing, leading to faster loan growth

Fed Funds Target Rate: Falling rate reduces borrowing costs and potentially encourages new borrowing, leading to faster growth

FTSE Bursa Malaysia KLCI: Rising stock market shows investor confidence and potentially easier access to capital, leading to faster loan growth

Correlation Analysis



Interpretations

- There is no high correlation among the factors but there is a moderate/weak correlation contribute to the growth.
- Most of the factor is positively correlated and only portions are negatively correlated that is fed funds target, FX and USD index.
- The correlation indicate that EXIM Bank asset growth had been influenced mostly by internal factors compared to external factors.
- Historically against the loan, the number of staff is the highest factor contributing to our growth and some main factor to the loan growth are increase of disbursed accounts, increase in staff strength and the strengthen of US GDP.
- number of disbursed accounts need to improve to increase number of application.
- Front office is more important in loan growth more than back office.
- MY market has small impact to the business compared to US market.

Multivariate Analysis

Growth Model

- Out of 10 variable obtained, only 9 is shown as significance while 1 variable (Disbursed Accounts) is removed to strengthen forecast model of loan projection in deciding the direction of whether loan growth.
- Model shows 98% accuracy and reliable for projections.
- Variable and factor are listed as per below:

Variable	Factor
No. of Application	Internal
Staff Strength	Internal
Fed Funds Target	External
GDP (MY & US)	External
FX (USD to MYR)	External
FTSE Bursa Malaysia KLCI	External
OPR	External
US Dollar Index	External

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              YoY Growth   R-squared:                       0.980
Model:                             OLS   Adj. R-squared:                  0.888
Method:                  Least Squares   F-statistic:                     10.67
Date:                 Tue, 05 Mar 2024   Prob (F-statistic):             0.0886
Time:                         01:17:44   Log-Likelihood:                -15.264
No. Observations:                   12   AIC:                             50.53
Df Residuals:                        2   BIC:                             55.38
Df Model:                            9
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       -576.5177    131.759     -4.376      0.048   -1143.433      -9.603
Application (DF2)    -0.9771      0.143     -6.853      0.021      -1.591      -0.364
Fed Funds      8.9692      2.418      3.709      0.066      -1.434      19.373
MY GDP         0.4417      0.198      2.229      0.156      -0.411       1.295
US GDP (DF2)   -1.1220      0.168     -6.661      0.022      -1.847      -0.397
FX (DF)        39.3279     11.802      3.332      0.079     -11.450      90.106
Bursa          0.3683      0.084      4.381      0.048       0.007       0.730
...
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.16e+05. This might indicate that there are
strong multicollinearity or other numerical problems.
```

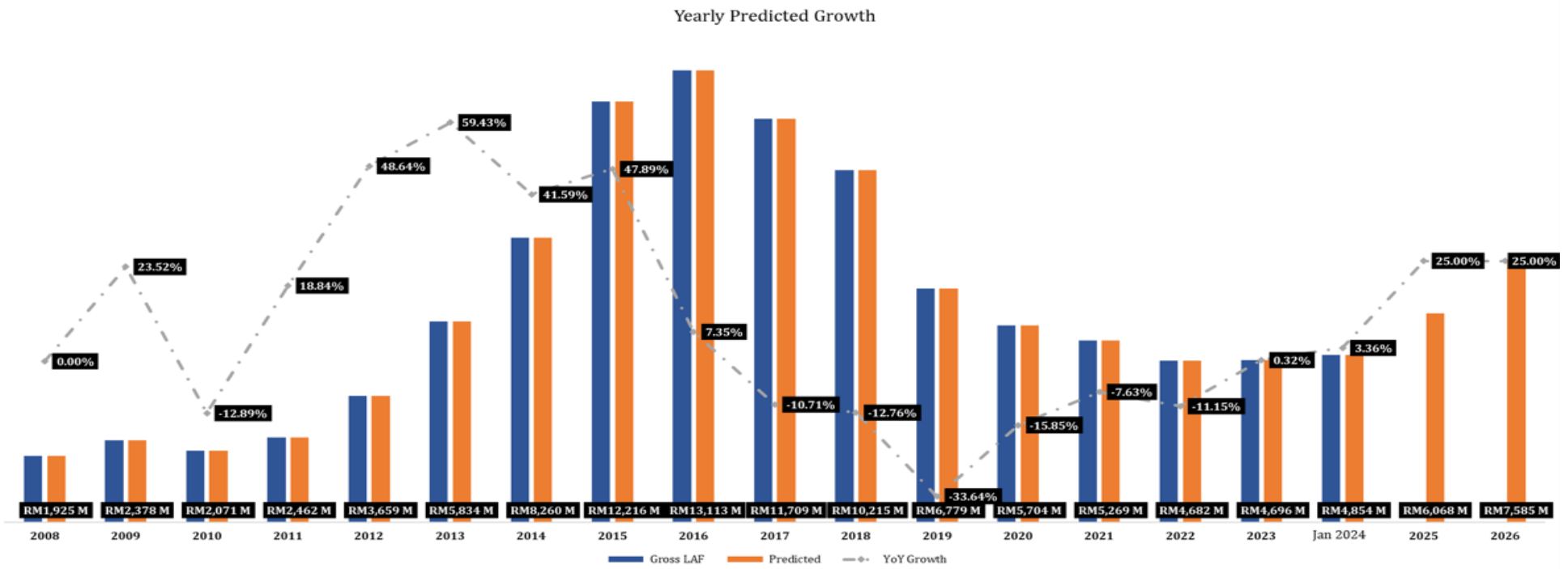
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Projection

Projections

- The model indicates that the optimum control to grow is the increase in manpower and business while the strengthen of cross border economy with low cost to the Malaysian country.
- The situation is very unlikely to control because it is expected to negate the requirement except the internal factors such as manpower and no. of application that can be work on to meet the requirement.
- Though the Malaysian Bank Negara Economic & Monetary Review expect the economy to expand between 4.0% to 5.0%, the global economic is potentially slowdowns in major economies, particularly China that could impact Malaysia's export-oriented industries and the Global Geopolitical tensions is uncertain that could disrupt the global trade and supply chains.
- The US economy's rising of consumer debt and elevated interest rates is as well could potentially slow down the economic growth particularly in consumer spending.

Variable	Current	Requirement	Transition
No. of Application	53	63	↑
Staff Strength	324	346	↑
Fed Funds Target	5.25	4.27	↓
GDP (Malaysia)	3.40	-16.60	↓
GDP (US)	3.10	10.97	↑
FX (USD to MYR)	4.70	4.47	↓
FTSE Bursa Malaysia KLCI	1447.04	1423.06	↓
OPR	3.00	2.84	↓
US Dollar Index	104.53	111.19	↑



Conclusion

Recommendation

- The model indicates realistic number for EXIM bank to achieve and the effectiveness is likely and reliable with high accuracy of 98% and the internal factors such as manpower and no. of application can be work on to meet the requirement.
- Main factors that driven the loan to grow is to work on the internal factors and the strengthen of US GDP, but it is unlikely for US to strengthen rapidly rather it will grow slowly due to effects of monetary policy, larger fiscal boost, dropped in housing sector activity, supply chain bottlenecks and geopolitical risk. The US economies also has rising of consumer debt and elevated interest rates is as well could potentially slow down the economic growth particularly in consumer spending.
- Though the Malaysian Bank Negara Economic & Monetary Review expect the economy to expand between 4.0% to 5.0%, the global economic is potentially slowdowns in major economies, particularly China that could impact Malaysia's export-oriented industries and the Global Geopolitical tensions is uncertain that could disrupt the global trade and supply chains.
- Historically against the loan, increase the number of staff has shown growth in loan mainly from front office. number of disbursed accounts need to improve to increase number of application.

Thank You